

Helix OTA â€” Fixed (closed workable items)

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This is the canonical closed-archive tracker (Â§11.4.19 column alignment, Â§11.4.33 type-aware closure vocabulary, Â§11.4.54 OTA-NNN). Open items live in [Issues.md](#); the short-form companion is [Fixed Summary.md](#). Sorted closure-date DESC. All commit hashes below were read from `git log`, not invented.

Â§1. [OTA-001] Telemetry deployment_id not derivable from otaprotocol.UpdateAvailable

Status: Fixed (â†’ Fixed.md) Type: Bug

Closed 2026-06-10 â€” commit 3c57867 feat(api): close 2 protocol gaps â€” UpdateAvailable.deployment_id + GET /deployments. The ota-protocol UpdateAvailable payload now carries the deployment_id field, wired through the serverâ€™s update-check response, so a real device can obtain and echo back the deployment_id its telemetry must supply (previously the server required it but the protocol never told the device). Proven by TestEmulatorSelfServesDeploymentID â€” the Tier-1 emulator round-trips register â†’ update-check â†’ telemetry self-serving the deployment_id end-to-end.

Â§2. [OTA-002] No GET /deployments list endpoint to enumerate deployments

Status: Implemented (â†’ Fixed.md) Type: Feature

Closed 2026-06-10 â€” commit 3c57867 feat(api): close 2 protocol gaps â€” UpdateAvailable.deployment_id + GET /deployments. A GET /deployments list endpoint now enumerates existing deployments with the same cursor-pagination convention as the group/members endpoints (50ef5c6), so the dashboard and operator/automation clients can discover deployment IDs without already knowing them. Proven by TestDeploymentListReturnsActive â€” the list endpoint returns the active deployments.

Â§5. [OTA-005] Per-device telemetry filters (?event / ?since / ?until)

Status: Implemented (â†’ Fixed.md) Type: Feature

Closed 2026-06-10 â€” commit 50ef5c6 feat(api): per-device telemetry filters + group/members pagination. Per-device telemetry endpoint now accepts ?event, ?since, ?until query filters; OpenAPI synced (redocly-clean).

Â§6. [OTA-006] Group + group-members cursor pagination

Status: Implemented (â†’ Fixed.md) **Type:** Feature

Closed 2026-06-10 â€” commit 50ef5c6 feat(api): per-device telemetry filters + group/members pagination. Group and group-members listing endpoints gained cursor-based pagination.

Â§7. [OTA-007] Dashboard API client synced to new pagination/filter params

Status: Completed (â†’ Fixed.md) **Type:** Task

Closed 2026-06-10 â€” commit b0b8ee2 chore(dashboard): sync API client to new pagination/filter params. Dashboard API client brought into lockstep with the new telemetry-filter + pagination parameters (OTA-005/OTA-006).

Â§8. [OTA-008] Tier-1 Go OTA device-emulator

Status: Implemented (â†’ Fixed.md) **Type:** Feature

Closed 2026-06-10 â€” commit 7dc3334 feat(emulator): Tier-1 Go OTA device-emulator + resilience for new handlers. A Go device-emulator (server/internal/deviceemu + cmd/ota-device-emu) that speaks the real ota-protocol to the control plane; surfaced the telemetry deployment_id protocol gap now tracked as OTA-001.

Â§9. [OTA-009] Comprehensive dashboard UI testing system

Status: Completed (â†’ Fixed.md) **Type:** Task

Closed 2026-06-10 â€” commit fa571b8 test(dashboard): comprehensive UI testing system for all panels. UI test suite covering all dashboard panels (Vitest + Playwright + ally).

Â§10. [OTA-010] Autonomous e2e + security + HelixQA for telemetry filters & pagination

Status: Completed (â†’ Fixed.md) **Type:** Task

Closed 2026-06-10 â€” commit a839220 test(qa): autonomous e2e + security + HelixQA for telemetry filters & pagination. Autonomous end-to-end + security suites + HelixQA bank covering the OTA-005/OTA-006 telemetry-filter and pagination work.

Â§11. [OTA-015] A/B slot switch via U-Boot BOOT_ORDER

Status: Implemented (â†’ Fixed.md) **Type:** Feature

Closed 2026-06-11 â€” commit 18ed84a feat(AB): PWU-AB-1 A/B slot switch proven. Full A/B slot switch via U-Boot BOOT_ORDER env var against real U-Boot 2024.01 + QEMU virt + HVF on the emulator T1 ladder. Boots A â†’ reboots to B via fw_setenv â†’ verifies HELIX_SLOTID=B, HELIX_ROOTDEV=/dev/vda3. Proven 3/3 deterministic. Evidence: docs/qa/20260611T094958Z-ab-slot-switch/. Runtime signature registered in docs/design/rk3588_ab_virt/runtime-signatures.yaml (PWU-AB-1-SLOT-SWITCH).

Â§12. [OTA-016] RAUC dd-apply to inactive slot with dm-verity

Status: Implemented (â†’ Fixed.md) **Type:** Feature

Closed 2026-06-20 â€” PWU-AB-2 RAUC reconciliation: dd rootfs to inactive slot + fw_setenv BOOT_ORDER switch so guest boots the new slot. RAUC slot-class scheme reconciled with uboot_ab/boot.cmd BOOT_ORDER env scheme. Proven 3/3 deterministic. Evidence: docs/qa/20260620T051026Z-ab-rauc-verity/. Runtime signature: PWU-AB-2-RAUC-DD.

Â§13. [OTA-017] U-Boot corrupt-slot auto-rollback via bootcount

Status: Implemented (â†’ Fixed.md) **Type:** Feature

Closed 2026-06-11 â€” commit 42be557 feat(AB): PWU-AB-3 corrupt-slot AUTO-ROLLBACK proven. bootcount > bootlimit triggers altbootcmd which swaps to the good slot. ROLLBACK run booted from bad slot B and rolled back to good slot A (HELIX_SLOTID=A, HELIX_ROOTDEV=/dev/vda2). Proven 2/2 deterministic. Evidence: docs/qa/20260611T095918Z-ab-rollback/. Runtime signature: PWU-AB-3-ROLLBACK.

Â§14. [OTA-018] ApplyPort CLI + slot manager + Ed25519 verifier

Status: Implemented (â†’ Fixed.md) **Type:** Feature

Closed 2026-06-20 â€” PWU-AB-4 ApplyPort (server/cmd/applyport/). Slot detection from /proc/cmdline (helix_slot=A|B), Ed25519 artifact signature verification via crypto/ed25519 (proven real, not a stub â€” Â§1.1 mutation TestMutationSignatureUsesRealEd25519), write-and-arm with health marker, and device client (login, check-for-update, apply). 58/58 tests passing including Â§1.1 paired-mutation suite. Evidence: server/internal/device/. Runtime signatures: PWU-AB-4-

Â§9. [OTA-019] Build resource stats tracker

Status: Completed (â†’ Fixed.md) **Type:** Task

Closed: 2026-06-20 â€” commit 300351c6 feat(stats): OTA-002 Build resource stats tracker. **Root cause:** Â§11.4.24 mandated per-build resource telemetry; missing entirely. **Fix:** scripts/resource_sampler.sh â€” background-capable host-side sampler (memory RSS, CPU%, load, disk I/O at 5s intervals, min/max/mean/p95). docs/Stats.tsv TSV registry. docs/Stats.md human-readable report. Tested end-to-end (10s run produces correct TSV + Stats.md). Observer constraint verified: <50MB RSS, <5% CPU. **Evidence:** docs/Stats.tsv, docs/Stats.md, tests/stats_resource_sampler_test.sh.

Â§15. [OTA-014] Docs Chain submodule distribution (Â§11.4.106 Phase 6)

Status: Completed (â†’ Fixed.md) **Type:** Task

Closed: 2026-06-20 **Root cause:** Docs Chain engine was used in-tree but not registered as a git submodule (Â§11.4.106 Phase 6). **Fix:** Registered vasic-digital/docs_chain@2c8cf16 as submodule. Added helix-deps.yaml entry. Fixed 2 context YAMLS (features-status fingerprint block, helixtrack inline transforms). All 4 contexts pass doctor. **Evidence:** .gitmodules, helix-deps.yaml, docs_chain doctor --all (4/4 PASS).

Â§16. [OTA-020] Database migration test

Status: Fixed (â†’ Fixed.md) **Type:** Bug

Closed: 2026-06-21 **Root cause:** workable_items.db had no automated validation for schema (tables, columns, constraints) or sync consistency with Issues.md/Fixed.md. **Fix:** Created scripts/testing/test_workable_items_db.sh â€” 47 tests covering schema validation, Issues sync (4/4), Fixed sync (12/12), orphan detection, terminal status checks. All 47/47 PASS. **Evidence:** qa-results/workable_items/20260620T201912Z/, scripts/testing/test_workable_items_db.sh, scripts/testing/migrate_workable_items.sh.